Subject: Data Structures
Code: CS200
Name: .....
Section: .....

Mid-Term Exam
Model No. A
Time: 60 minutes
Full Marks: 15 Marks

Question 1:

(6 Marks)

a) Complete the table below to compare and differentiate between the following data structures.

Data Structure	Underlying Structure	Order of Processing (FIFO/LIFO/Priority)	One Common Use Case
Stack	Array or Linked List	LIFO	Web Navigation (Back / Forward)
Queue	Array or Linked List	FIFO	Customer service systems
Priority Queue	Heap	Priority	CPU scheduling

b) Why is a Linked List often a good choice for implementing both a Stack and a Queue?

Because a linked list allows dynamic memory allocation and makes insertion and deletion operations easy and efficient without shifting elements, which is suitable for both stack and queue operations.

Question 2:

(3 Marks)

Given the following Node structure:

```
struct Node {
    int data;
    Node* next;     };
```

Write a C++ function called **countNodes** that takes the head pointer of a singly linked list and returns the number of nodes in the list.

Function Signature:

```
int countNodes(Node* head);
```

```
int countNodes(Node* head)
{
    int count = 0;
    Node* current = head;
    while(current != NULL)
    {
        count++;
        current = current->next;
    }
    return count;
}
```

Question 3: **(6 Marks)**

Write a C++ function that simulates a simple customer service system using a queue. The function should: *(Note: **Don't use the STL** to implement the queue operations.)*

1. Add the following customers to the queue: "Alice", "Bob", "Charlie"
2. Serve the next customer (remove from front)
3. Add a new customer "David" to the queue
4. Return the name of the customer who is currently at the front of the queue

Function Signature:

```
#include <string>
using namespace std;

string simulateCustomerService();
```

```
#include <iostream>
using namespace std;

string simulateCustomerService()
{
    string queue[10];
    int front = 0;
    int rear = -1;
Q1
    rear++;
    queue[rear] = "Alice";

    rear++;
    queue[rear] = "Bob";

    rear++;
    queue[rear] = "Charlie";

Q2
    front++;

Q3
    rear++;
    queue[rear] = "David";

Q4
    return queue[front];
}
```

WITH MY BEST WISHES

solved by Eng Adham Elboghdady